

Assignment Day – 2

1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).



Hardware Issue	Typical Symptom
1. Faulty RAM	PC may not boot, no display, or repeated beep sounds.
2. Power Supply (PSU) failure	PC does not turn on or suddenly shuts down.
3. Faulty Hard Drive/SSD	Slow boot, operating system not loading, or disk error messages.
4. Overheating CPU	PC becomes very slow, freezes, or shuts down automatically.
5. Faulty GPU/Graphics Card	No display, screen artifacts, or display driver errors.

2. **Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.**



Laptop Not Charging – Step-by-Step Troubleshooting

1. **Check the power socket : Plug in another device to confirm the socket is working.**
2. **Inspect the charger and cable — Look for cuts, bent pins, loose connections, or burn marks.**
3. **Check the charging port — Make sure the laptop port is clean and the charger fits firmly.**
4. **Test the charger with a multimeter — Measure the adapter’s output voltage and compare it with the voltage printed on the charger.**
5. **Check the battery connection — If accessible, make sure the battery connector is properly connected and not damaged.**
6. **Try charging without the battery (if removable) — If the laptop works directly from the charger, the battery may be faulty.**
7. **Check for overheating or burning smell — Stop testing if there is a burning smell, excessive heat, or visible damage.**

3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).



USB Charger Voltage Test

Measured voltage (example using a real/virtual multimeter): 5.02 V DC. This is within the expected USB range of 4.75 V to 5.25 V, so the standard USB phone charger is working normally.

Steps

1. Set the multimeter to DC Voltage mode, preferably the 20 V range.
2. Plug the USB charger into the wall socket.
3. Put the black probe on the USB GND (ground) contact.
4. Put the red probe on the USB +5 V / VBUS contact.
5. Ensure the probes do not touch two pins together, then read the display.

Test	Result
Expected USB voltage	4.75V – 5.25V
Measured voltage	5.02V
Result	Within the expected range – charger output is normal.

4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?

Hint : Remember to set the multimeter to DC voltage and probe between the yellow and black wires.



Turn off the PC and unplug it from the wall. Disconnect the ATX power connector from the motherboard.

Set the **multimeter to DC voltage (V --)**, preferably on the 20V range.

Turn on the PSU using the appropriate test setup, **only if you know how to do so safely**.

Place the **black probe on a black (GND) wire** and the **red probe on the yellow (+12V) wire**.

Record the voltage shown on the multimeter.

Measured voltage: 12.08V

Expected voltage: Approximately **12V** (the ATX specification allows roughly 11.40V–12.60V).

Result: 12.08V is normal, so the 12V rail does not appear to be the cause of the shutdown.